Electrodynamics
Worksheet PS.10

(Chapter 3)
December 8, 2025
Zoeb Izzi

1. Applying quantum ideas to kinematics

Suppose someone drops a rock off a cliff of height h. As it falls, you decide to snap a million pictures at random intervals. In each image, you measure the distance the rock has fallen.

a. What is the position of the rock as a function of time? Use your AP Physics skills and only employ *classical* kinematics. For a reference frame, set the ground as $y = 0$ and upward as the positive y direction.

   Trivially, we know that it is $\boxed{y = h - \frac{1}{2}gt^2}$.

b. What is the velocity of the rock as a function of time? Once again, *only* use classical kinematics for this calculation.

   Once again, we trivially deduce that $\boxed{v = -gt}$.

c. What is the total time the rock is in the air?

$$0 = h - \frac{1}{2}gt^2, \boxed{t = \sqrt{\frac{2h}{g}}}$$

d. Determine an expression for the probability density in terms of height (y) for the ball as it falls.

$$t = \sqrt{\frac{2(h-y)}{g}} \implies \left|\frac{\mathrm{d}t}{\mathrm{d}y}\right| = \frac{1}{\sqrt{2g(h-y)}} \implies \boxed{\rho(y) = \frac{1}{2\sqrt{h(h-y)}}, \quad 0 \le y \le h}$$

e. Confirm that your probability density correctly predicts that the rock MUST be found somewhere between $y = 0$ and $y = h$.

$$\int_0^h \rho(y)\,\mathrm{d}y = \frac{1}{2\sqrt{h}}\int_0^h (h-y)^{-1/2}\,\mathrm{d}y = \frac{1}{2\sqrt{h}}\Big[-2\sqrt{h-y}\Big]_0^h = \frac{1}{2\sqrt{h}} \cdot 2\sqrt{h} = 1 \checkmark$$

   Since the probability that it lies in this interval is 1, it must be the case that the probability it lies outside this interval is $1 - 1 = 0$, so it cannot lie outside the specified interval.

f. Use your probability density to calculate the expectation value of the position of the ball over the course of its entire flight. Why is this expectation value greater than $\frac{h}{2}$?

$$\langle y\rangle = \int_0^h y\,\rho(y)\,\mathrm{d}y = \frac{1}{2\sqrt{h}}\int_0^h \frac{y}{\sqrt{h-y}}\,\mathrm{d}y = \frac{1}{2\sqrt{h}}\int_0^h \frac{h-u}{\sqrt{u}}\,\mathrm{d}u = \frac{1}{2\sqrt{h}}\left[h\cdot 2\sqrt{h} - \frac{2}{3}h^{3/2}\right] = \frac{1}{2\sqrt{h}}\cdot\frac{4}{3}h^{3/2} = \boxed{\frac{2h}{3}}$$

   It makes sense that this value is greater than $\frac{h}{2}$ since we have an acceleration, which means it moves faster (thereby spending less time) when closer to the ground, so it would spend a larger percentage of it's time slightly higher - in this case, 33% higher.

2. Normalizing a wavefunction in the position basis

A particle is represented by the wave function:

$$\phi(x) = \begin{cases} A(x/a) & 0 \leq x \leq a \\ A(b-x)/(b-a) & a \leq x \leq b \\ 0 & otherwise \end{cases}$$

Where $A, a,$ and b are positive constants.

a. Normalize $\phi(x)$ (i.e. find A in terms of a and b.)

$\int_0^a A^2\left(\frac{x}{a}\right)^2$−rf /Library/TinyTeX/tlpkg/texlive.tlpdb∗tlmgroptionrepositoryhttps : //ctan.org/tex−archive/systems/

$$\int_a^b A^2 \frac{(b-x)^2}{(b-a)^2}\,\mathrm{d}x = \frac{A^2}{(b-a)^2} \cdot \frac{(b-a)^3}{3} = \frac{A^2(b-a)}{3}$$

$$(1) + (2) = \frac{A^2 b}{3} = 1 \implies \boxed{A = \sqrt{\frac{3}{b}}}$$

b. Sketch $\phi(x)$ as a function of x.

$x\psi(x)0ab0A$

c. Where is the particle most likely to be found?

Trivially, from the graph, at $x = a$.

d. What is the probability of finding the particle to the left of a? Check your result in the limiting cases, $b = a, 2a$.

e. If many measurements of the position (x) of this particle were made, what would be the average location the particle was found in?

3. Expectation Values & Uncertainty

A particle is represented (at time $t = 0$) by the wave function:

$$\phi(x) = \begin{cases} A(a^2 - x^2) & -a \leq x \leq a \\ 0 & otherwise \end{cases}$$

a. Determine the normalization constant A.

b. What is the expectation value of x?

c. What is the expectation value of p?

d. Find the expectation value of x^2.

e. Find the expectation value of p^2.

f. Find the uncertainty in x (σ_x).

g. Find the uncertainty in p (σ_p).

h. Check that your results are consistent with the uncertainty principle.

4. Infinite Square Well

A particle in the infinite square well has the initial wave function $\psi(x) = Ax(a-x)$ for $(0 \le x \le a)$. Outside of this domain, the wave function is, of course, $\psi(x) = 0$.

a. Find the normalization constant A.

b. We know we can always decompose any initial state into a superposition of eigenstates. We can write out this superposition as:

$$\psi(x) = \sum_{n=1} c_n \phi(n)$$

where we can determine the expansion coefficients using the overlap integral:

$$c_n = \int_0^a \phi(x) \cdot \psi(x) dx$$

with the eigenfunctions

$$\phi_n = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a} x\right)$$

Determine expressions for the values of c_n for $n =$ even integers and odd integers.

c. Make a plot (using Python, Desmos, etc) of $\psi(x)$ (as initially defined) and your expansion for $\psi(x) = \sum_{n=1} c_n \phi_n$. Let $a = 1$, and use only $n = 1, 2, 3$. How well do these two functions match?

d. What is the probability that the wave function $\psi(s)$ gets measured to be in the ground state $n = 1$?